Chapter 87 — Error Classification & Telemetry Pipeline

> **Part X — Performance, Observability & Quality** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 82 (analytics/logging pipeline), Ch 83 (crash/ANR), Ch 84 (tracing), Ch 85 (perf), Ch 28 (privacy), Ch 23 (error handling/Result). Pairs with Ch 81 (feedback/ diagnostics), Ch 67 (network errors).
>
> **Reference surface in StreamCart:** the unifying **telemetry pipeline** that ingests all signals — **errors** (handled + unhandled, Ch 83), **logs**, **metrics** (Ch 85), **traces** (Ch 84), **analytics** (Ch 82) — and the **error classification** scheme that turns raw failures into **actionable, grouped, severity-ranked, PII-safe** signal: fatal vs non-fatal, transient vs permanent, by domain/cause, **fingerprinted/deduplicated**, routed by severity.

87.1 Problem Definition and Real-World Context

Previous chapters built individual observability pillars — **analytics** (Ch 82), **crash/ANR** (Ch 83), **tracing** (Ch 84), **performance** (Ch 85). This chapter is the **unifying layer**: a **telemetry pipeline** that ingests **all** signals (errors, logs, metrics, traces, events) through one **reliable, batched, consented, PII-safe** path, and an **error classification** scheme that turns the flood of raw failures into **actionable** signal — **grouped** (fingerprinted/deduplicated so 10,000 occurrences of one bug are one issue), **classified** (fatal vs non-fatal, transient vs permanent, by domain), **severity-ranked** (what needs a page vs a dashboard), and **routed** accordingly. Without classification, telemetry is **noise**: you can't tell a retryable network blip (Ch 67) from a data-corruption bug, or a one-off from a crash-looping release (Ch 83).

Concretely, teams hit:

- **Signal silos.** Analytics, crashes, traces, metrics in **separate** SDKs/pipelines with no **correlation** (Ch 84) — you can't connect a slow trace to the crash it caused. Need a **unified pipeline** (or at least correlated via ids, Ch 84).
- **Unclassified errors = noise.** 50,000 errors/day with no **grouping** or **severity** is useless. Errors must be **classified**: **fatal** (crash, Ch 83) vs **non-fatal** (handled, Ch 23); **transient** (retryable network, Ch 67) vs **permanent** (bug/data); by **domain** (network/db/UI/payment). And **fingerprinted** so duplicates collapse.
- **No deduplication/fingerprinting.** The **same** bug reported 10,000 times should be **one issue** with a count — group by a **stable fingerprint** (stack signature, error type + location), not raw text (Ch 83).
- **No severity routing.** A **fatal, high-volume, new-in-this-release** error needs a **page** (Ch 83); a rare handled warning needs a **dashboard**. Severity drives **routing/alerting**.
- **PII in telemetry.** Errors/logs often carry PII (user data in a stack message, a URL with a token) — must be **redacted** (Ch 28) at the boundary.
- **Reliability/cost of the pipeline.** Telemetry must be **durably buffered, batched, sampled** (Ch 82) — not lose data on crash, not drain battery/data, not cost a fortune at volume.
- **Transient vs permanent handling.** A **transient** error (timeout, 503, Ch 67) is **retryable** and often **expected** (lower severity); a **permanent** one (400, parse failure, assertion) is a **bug** (higher severity). Classification

drives **handling** (retry vs report, Ch 23/67) and **severity**.

The architecture: a **unified telemetry pipeline** ingesting **errors/logs/metrics/traces/ events** (Ch 82/83/84/85) through one **reliable, batched, consented, PII-safe** path; **classify** each error — **fatal/non-fatal**, **transient/permanent**, by **domain** — **fingerprint and deduplicate** (group by stable signature, Ch 83), **rank by severity**, and **route** (page/dashboard/sample); **correlate** across pillars via ids (Ch 84); **redact PII** at the boundary (Ch 28); and **sample/buffer** for reliability + cost (Ch 82). This chapter is about **turning raw failure signal into actionable, deduplicated, severity-ranked, PII-safe telemetry**.

Three practical use cases

1. **Classified, deduplicated errors (StreamCart).** A null-deref in checkout (Ch 75) is **fingerprinted** (type + location) so its 8,000 occurrences are **one issue** with a count, **fatal/permanent** severity, routed to a **page** if new-in-release (Ch 83).
2. **Transient vs permanent network errors.** A 503 (Ch 67) is **transient/retryable**, logged at low severity and **sampled**; a 400 with a malformed request is **permanent**, surfaced as a bug.
3. **Cross-pillar correlation.** A slow trace (Ch 84/85) that ends in a handled error (Ch 23) is **correlated** by trace/correlation id so you see the full story (Ch 84).

Why this is architectural

This is the **capstone** of observability: it composes analytics (Ch 82), crash (Ch 83), tracing (Ch 84), performance (Ch 85), error handling (Ch 23/67), and privacy (Ch 28) into one **classified, deduplicated, correlated, PII-safe** telemetry system. The decisions — classification scheme, fingerprinting/dedup, severity routing, unified+correlated ingestion — determine whether you can **operate** the app from signal rather than drown in noise.

87.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Ingest errors/logs/metrics/traces/events through one reliable, batched pipeline (Ch 82/83/84/85).
F2	Classify errors: fatal/non-fatal, transient/permanent, by domain.
F3	Fingerprint + deduplicate (group by stable signature, Ch 83).
F4	Rank by severity; route (page/dashboard/sample) (Ch 83).
F5	Correlate across pillars via ids (Ch 84).
F6	Redact PII at the boundary; consent (Ch 28); buffer/sample for reliability+cost (Ch 82).

Non-functional requirements

Attribute	Policy
Actionability	Classified, grouped, severity-ranked — not noise.
Reliability	Durably buffered; survives crash (Ch 82/83).
Privacy	PII redacted; consent (Ch 28).
Cost	Batched + sampled by severity (Ch 82).
Correlation	Cross-pillar via ids (Ch 84).

Scope

- **In scope:** unified telemetry ingestion (errors/logs/metrics/traces/events, Ch 82/83/84/85), error classification (fatal/non-fatal, transient/permanent, domain), fingerprinting/dedup (Ch 83), severity ranking + routing (page/dashboard/sample), cross-pillar correlation (Ch 84), PII redaction + consent (Ch 28), buffering/sampling (Ch 82), transient-vs-permanent handling (Ch 23/67).
- **Out of scope:** analytics internals (Ch 82), crash capture mechanics (Ch 83), tracing propagation (Ch 84), perf metrics (Ch 85), error-handling patterns (Ch 23), network retry (Ch 67).

Assumptions & constraints

- A telemetry/observability backend (Crashlytics + analytics + tracing, or a unified APM) reached via the reliable pipeline (Ch 82). Errors **classified + fingerprinted**; severity drives **routing**; signals **correlated** by id (Ch 84); PII **redacted** (Ch 28); data **buffered + sampled**. `minSdk 24`.

87.3 System Design Fundamentals

- **Telemetry pillars, unified (Ch 82/83/84/85).** The signals: **logs** (discrete events), **metrics** (aggregated numbers, Ch 85), **traces** (causal spans, Ch 84), **events** (analytics, Ch 82), **errors/crashes** (Ch 83). A mature system ingests them through **one reliable, batched, consented, PII-safe pipeline** (Ch 82) and **correlates** them via shared ids (Ch 84) — so a slow trace, its metrics, and the error it produced tell one story.
- **Error classification (the core taxonomy):**
  - **Fatal vs non-fatal: fatal** = crash/ANR (process died, Ch 83); **non-fatal** = handled error the app recovered from (Ch 23) but is worth tracking. Different severity, different handling.
  - **Transient vs permanent: transient** = retryable, often expected (timeout, 503, offline, Ch 67/67) — lower severity, **sampled**; **permanent** = a bug or bad data (400, parse failure, `IllegalStateException`, assertion) — higher severity, surfaced. This drives **handling** (retry vs report, Ch 23) **and** severity.
  - **By domain/cause:** network (Ch 67), database (Ch 5), UI/render (Ch 90), payment (Ch 76), auth (Ch 20) — so you route to the right team and dashboard.
- **Fingerprinting & deduplication (Ch 83).** The **same** error happening 10,000 times is **one issue**, not 10,000 — compute a **stable fingerprint** (exception type + top app frame/ location, normalized message) and **group** by it, keeping a **count + first/last seen + affected users + versions** (Ch 83). Group by **signature**, not raw text (which varies by dynamic values). Good fingerprinting is what makes error tools usable.

- **Severity ranking + routing.** Rank by **impact**: **fatal + high-volume + new-in-this- release + affecting many users** = **page** (Ch 83); a rare handled warning = **dashboard only**; transient/expected = **sampled/aggregated**. Severity drives **alerting** (Ch 83) and **sampling** (keep all fatals, sample transients, Ch 82). Velocity (spiking) matters as much as volume.
- **Correlation across pillars (Ch 84).** Attach **trace/correlation/session ids** (Ch 84) to every signal so an error links to its **trace** (what led to it, Ch 84), its **metrics** (Ch 85), the **session/release/device** context — turning an isolated error into an **investigable** story (Ch 84/81).
- **PII redaction at the boundary (Ch 28).** Errors/logs are the **biggest PII leak risk** — a stack message with user data, a URL with a token, a log line with an email. **Redact** at the pipeline boundary (allowlist fields, scrub patterns, Ch 28) **before** upload. Respect **consent** (Ch 82).
- **Reliability + cost (Ch 82).** The pipeline is **durably buffered** (survives crash — the crash report itself must persist, Ch 83), **batched**, and **sampled by severity** (keep all fatals, sample high-volume transients) to bound **battery/data/cost** (Ch 82). Telemetry mustn't lose the important signal or bankrupt you on the noise.
- **Structured, not free-text.** Signals are **structured** (type, severity, domain, fingerprint, context, ids) — queryable and groupable — not free-text logs. Structure is what enables classification, dedup, and correlation.

Core insight: **unify the observability pillars — logs, metrics (Ch 85), traces (Ch 84), analytics events (Ch 82), errors/crashes (Ch 83) — into one reliable, batched, consented, PII-safe telemetry pipeline, and classify every error along three axes (fatal vs non-fatal, transient/retryable vs permanent/bug, and by domain) so classification drives both handling (retry vs report, Ch 23/67) and severity; fingerprint and deduplicate errors by a stable signature (type + location) so thousands of occurrences collapse into one counted, ranked issue (Ch 83); rank by severity and velocity and route accordingly (page the fatal, new-in-release, high-impact ones; dashboard the rest; sample the expected transients, Ch 82); correlate every signal across pillars by trace/correlation/session ids (Ch 84) so an error links to the trace, metrics, and context that explain it; and redact PII at the boundary with consent (Ch 28) while durably buffering and severity-sampling for reliability and cost (Ch 82) — turning a flood of raw failures into actionable, deduplicated, correlated, PII-safe signal.**

87.4 High-Level Architecture


87.4 High-Level Architecture

Components & ownership

- **Classifier** tags each error (fatal/non-fatal, transient/permanent, domain).
- **Fingerprinter** groups/deduplicates by stable signature (Ch 83).
- **Severity + router** ranks and routes (page/dashboard/sample, Ch 83/82).
- **Correlator** attaches ids (Ch 84); **Redactor** scrubs PII (Ch 28); **pipeline** buffers/ batches/samples (Ch 82).

Direction: signals (Ch 82/83/84/85) → classify → fingerprint/dedup → severity → route; correlated (Ch 84), redacted (Ch 28), buffered (Ch 82).

87.5 Detailed Component Design


87.5 Detailed Component Design

Component	Responsibility
`TelemetryEvent`	structured signal: type/severity/domain/fingerprint/transient/ids (Ch 84)
`ErrorClassifier`	fatal/non-fatal, transient/permanent, domain
`Fingerprinter`	stable signature for grouping/dedup (Ch 83)
`SeverityRouter`	rank + route (page/dashboard/sample, Ch 83/82)
`TelemetryPipeline`	redact (Ch 28) + buffer + batch + sample (Ch 82)

87.6 End-to-End Data Flow

Error → classify → fingerprint → severity → route



Transient vs permanent → handling + severity



87.7 Complete Kotlin Implementation

Package: `com.streamcart.telemetry`.

87.7.1 Classification (fatal/non-fatal, transient/permanent, domain)

```
data class TelemetryEvent(
    val type: String, val severity: Severity, val domain: Domain,
    val fingerprint: String, val transient: Boolean, val fatal: Boolean,
    val traceId: String?, val context: Map<String, String>,   // correlated (Ch 84), PII-safe
(Ch 28)
)
enum class Severity { FATAL, ERROR, WARN, INFO }
enum class Domain { NETWORK, DATABASE, UI, PAYMENT, AUTH, UNKNOWN }

@Singleton
class ErrorClassifier @Inject constructor(private val fingerprinter: Fingerprinter, private val
correlation: CorrelationManager) {  // Ch 84
    fun classify(t: Throwable, fatal: Boolean): TelemetryEvent {
        val transient = isTransient(t)                                      // retryable vs bug
        return TelemetryEvent(
            type = t::class.simpleName ?: "Unknown",
            severity = when { fatal -> Severity.FATAL; transient -> Severity.WARN; else ->
Severity.ERROR },  // transient -> lower
            domain = domainOf(t), fingerprint = fingerprinter.fingerprint(t),   // group/dedup
(Ch 83)
            transient = transient, fatal = fatal, traceId = correlation.currentTraceId(),   //
correlate (Ch 84)
            context = mapOf("release" to BuildConfig.VERSION_NAME))
    }
    private fun isTransient(t: Throwable): Boolean = when (t) {            // transient =
retryable (Ch 67)
        is SocketTimeoutException, is UnknownHostException, is IOException -> true
        is HttpException -> t.code() in listOf(429, 500, 502, 503, 504)   // 5xx/429 retryable
        else -> false                                                      // 4xx (except 429),
parse, state -> permanent (bug)
    }
    private fun domainOf(t: Throwable): Domain = when (t) {
        is IOException, is HttpException -> Domain.NETWORK                 // Ch 67
        is SQLException -> Domain.DATABASE                                 // Ch 5
        else -> Domain.UNKNOWN
    }
}
```

87.7.2 Fingerprinting / deduplication (Ch 83)

```
@Singleton
class Fingerprinter @Inject constructor() {
    /** Stable signature so thousands of occurrences group into ONE issue (Ch 83). */
    fun fingerprint(t: Throwable): String {
        val top = t.stackTrace.firstOrNull { it.className.startsWith("com.streamcart") }  // top
APP frame
        val sig =
"${t::class.qualifiedName}@${top?.className}.${top?.methodName}:${top?.lineNumber}"
        return MessageDigest.getInstance("SHA-256").digest(sig.toByteArray()).joinToString("") {
"%02x".format(it) }.take(16)
        // Group by TYPE + LOCATION, not raw message (dynamic values vary) - stable across
occurrences.
    }
}
```

87.7.3 Severity routing (page/dashboard/sample)

```
@Singleton
class SeverityRouter @Inject constructor() {
    enum class Route { PAGE, DASHBOARD, SAMPLE }
    /** Route by severity (backend also factors volume/velocity/new-in-release, Ch 83). */
    fun route(event: TelemetryEvent): Route = when {
        event.fatal -> Route.PAGE                      // fatal -> page (Ch 83)
        event.transient -> Route.SAMPLE                // transient/expected -> sample+aggregate
(Ch 82)
        else -> Route.DASHBOARD                         // permanent non-fatal -> dashboard
    }
    // Backend escalates DASHBOARD -> PAGE if volume spikes / new-in-release / many users (Ch
83).
}
```

87.7.4 Pipeline: redact → buffer → batch → sample (Ch 82/28)

```
@Singleton
class TelemetryPipeline @Inject constructor(
    private val redactor: PiiRedactor,    // Ch 28
    private val buffer: TelemetryBuffer,   // durable (survives crash, Ch 83)
    private val consent: ConsentManager,   // Ch 82
) {
    fun report(event: TelemetryEvent) {
        if (!consent.telemetryAllowed()) return                          // consent (Ch 82)
        val safe = redactor.redact(event)                                // scrub PII at
boundary (Ch 28)
        val keep = event.fatal || Random.nextFloat() < sampleRate(event) // keep ALL fatals;
sample transients (Ch 82)
        if (keep) buffer.enqueue(safe)                                    // durable buffer ->
batched upload (Ch 82)
    }
    private fun sampleRate(e: TelemetryEvent) = if (e.transient) 0.05f else 1.0f   // sample
expected transients
}
// PiiRedactor: allowlist context fields + scrub email/token/card patterns (Ch 28) BEFORE
upload.
```

87.7.5 Hooking it up (fatal + non-fatal)

```
// Fatal (crash): uncaught handler (Ch 83) -> classify(fatal=true) -> pipeline (persist before
death).
// Non-fatal (handled, Ch 23): in a catch, telemetry.report(classifier.classify(e, fatal =
false)).
// Transient network (Ch 67): classified transient -> sampled, low severity (expected).
class CheckoutRepo(private val telemetry: Telemetry) {
    suspend fun place(order: Order): Result<Receipt> = runCatching { api.place(order) }  // Ch
23
        .onFailure { telemetry.report(it, fatal = false) }   // classified, fingerprinted,
correlated (Ch 84)
}
```

87.7.6 Module structure

```
:core:telemetry  TelemetryEvent, ErrorClassifier, Fingerprinter, SeverityRouter,
TelemetryPipeline
# Ingests crash (Ch 83), traces (Ch 84), metrics (Ch 85), events (Ch 82); correlate -> Ch 84;
# redact -> Ch 28; buffer/batch/sample -> Ch 82; dedupe/rank/route -> backend. Handling -> Ch
23/67.
```

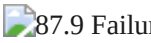
87.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Error flood	noise + cost	fingerprint/dedup + severity sampling (Ch 83/82)
Transient treated as bug	false alarms	classify transient -> low severity/sample (Ch 67)
Unstable fingerprint	duplicate issues	type + location signature (not raw message, Ch 83)
PII in error	leak	redact at boundary (Ch 28)
Crash loses its own report	no signal	durable buffer, persist-before-death (Ch 83)
No correlation	un-investigable	attach trace/correlation ids (Ch 84)
Everything pages	alert fatigue	severity routing (page fatal only, Ch 83)

- **Fingerprinting + severity sampling** is what keeps the pipeline **affordable and actionable** under a flood — group duplicates, keep all fatals, sample expected transients (Ch 83/82).
- **Classification drives handling** (Ch 23/67): a transient error is **retried** and logged quietly; a permanent one is **surfaced** — so the taxonomy isn't just reporting, it shapes behavior.
- **Durable buffering** ensures the **crash's own report survives** the crash (Ch 83) — the most important signal is the one most at risk.

87.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Error flood drowns signal	volume spike	fingerprint/dedup + severity sample (Ch 83/82)	actionable
Transient alarms as bug	false pages	classify transient (Ch 67)	quiet
Same bug = 10k issues	duplicate groups	stable fingerprint (Ch 83)	one issue
PII leaked in error	privacy audit	redact at boundary (Ch 28)	safe
Crash report lost	missing signal	durable buffer (Ch 83)	captured
Can't investigate error	no context	correlate ids (Ch 84)	investigable
Alert fatigue	too many pages	severity routing (Ch 83)	focused


87.9 Failure Scenarios and Recovery

> **Classify, fingerprint, rank, route — and redact.** The flood of raw errors becomes actionable only when you classify (fatal/transient/domain), group duplicates by a stable signature (Ch 83), rank by severity and velocity, and route (page the fatal, sample the transient, Ch 82). Correlate across pillars (Ch 84) to investigate, redact PII at the boundary (Ch 28), and durably buffer so the crash's own report survives (Ch 83).

87.10 Scaling the Implementation

- **Level 1.** Crashlytics (fatal + non-fatal, Ch 83) with default fingerprinting; analytics (Ch 82) separate.
- **Level 2.** Explicit classification (transient/permanent/domain); custom fingerprinting; severity routing; PII redaction (Ch 28).
- **Level 3 — production.** Unified pipeline (Ch 82) with correlation ids (Ch 84); severity + velocity routing (page/dashboard/sample); durable buffering; consent (Ch 28).
- **Level 4 — platform.** One telemetry backend (APM) ingesting logs/metrics/traces/errors/ events with full correlation (Ch 84); SLO-based alerting; automated grouping/anomaly detection.
- **Level 5 — enterprise.** Observability platform: unified schema, tiered sampling, cost governance, cross-service correlation (Ch 84), error budgets, on-call routing by domain/ ownership.

Avoid overengineering: Crashlytics + analytics with explicit transient/permanent classification, stable fingerprinting, severity routing, and PII redaction covers most apps. Don't page on transients; don't skip fingerprinting; don't leak PII; don't silo the pillars without correlation.

87.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Pipeline	Unified + correlated (Ch 84)	siloed SDKs	Unified/correlated (one story)
Grouping	Fingerprint/dedup (Ch 83)	raw, ungrouped	Fingerprint (actionable)
Classification	Transient/permanent/domain	unclassified	Classify (drives handling+severity)
Routing	Severity + velocity (Ch 83)	all-or-nothing	Severity routing (focus)
Sampling	By severity (keep fatals, Ch 82)	uniform/none	Severity sampling (cost)
PII	Redact at boundary (Ch 28)	raw	Redact (privacy)
Fingerprint key	Type + location	raw message	Type+location (stable)

Unified/correlated vs siloed: separate SDKs for analytics, crashes, traces, metrics mean you **can't connect** a slow trace to the error it caused (Ch 84). A **unified** (or id- correlated, Ch 84) pipeline tells **one story** — error ↔ trace ↔ metrics ↔ context. Prefer correlation (shared ids, Ch 84) even if backends differ.

Fingerprint by type+location vs raw message: grouping by **raw message** fails — dynamic values (ids, timestamps) make every occurrence look unique, so 10,000 instances become 10,000 issues. Grouping by **exception**

type + top app frame/location is **stable** across occurrences → one counted issue (Ch 83). Always fingerprint by signature.

Severity sampling vs uniform: sampling **everything uniformly** either drops important fatals or keeps too much noise. **Severity sampling** — **keep all fatals** (Ch 83), **sample expected transients** (Ch 82) — keeps the signal while bounding cost. Sample by severity.

87.12 Security & Privacy Considerations

- **Redact PII at the boundary (Ch 28):** errors/logs are the **top PII leak** — stack messages with user data, URLs with tokens, logged payloads. **Allowlist** context fields, **scrub** email/token/card patterns **before** upload (Ch 28). Never ship raw error text with PII.
- **Consent (Ch 82):** telemetry follows the consent regime; opted-out users limited to essential crash signal (minimal, anonymized).
- **Correlation ids opaque (Ch 84):** link signals by **opaque** ids, not PII.
- **No secrets in context:** release/device/screen in context, never tokens/keys/card data (Ch 76).
- **Access-controlled telemetry:** error/trace data is internal, access-controlled, retention- limited (Ch 28).
- Client vs backend: the **client** classifies, fingerprints, redacts, and buffers PII-safely; the **backend** dedupes, ranks, routes, and access-controls — neither stores PII in telemetry.

87.13 Performance and Resource Optimization

- **Classification/fingerprinting are cheap** — type/stack inspection + one hash; done off the hot path (in the reporting path, Ch 82).
- **Severity sampling bounds volume/cost** — keep fatals, sample transients (Ch 82).
- **Batched, buffered upload** (Ch 82) — no per-error network; survives crash (Ch 83).
- **Dedup reduces backend cost** — one issue per fingerprint, not per occurrence (Ch 83).
- **Redaction is bounded** — allowlist + pattern scrub, not heavy parsing (Ch 28).
- **Profiling:** reporting-path overhead (should be negligible), event volume by severity, sampling rates, buffer size, upload batch efficiency (Ch 82).

87.14 Testing Strategy

Classification, fingerprinting, routing, and redaction are all unit-testable.

```
class ErrorClassifierTest {
    @Test fun `timeout is transient and low severity`() {
        val e = classifier.classify(SocketTimeoutException(), fatal = false)
        assertTrue(e.transient); assertEquals(Severity.WARN, e.severity);
assertEquals(Domain.NETWORK, e.domain)  // Ch 67
    }
    @Test fun `parse error is permanent`() {
        assertFalse(classifier.classify(JsonParseException("x"), fatal = false).transient)   //
bug, higher severity
    }
}

class FingerprinterTest {
    @Test fun `same type+location groups despite different messages`() {
        val a = thrownAt("checkout", msg = "id=1"); val b = thrownAt("checkout", msg = "id=2")
        assertEquals(fingerprinter.fingerprint(a), fingerprinter.fingerprint(b))   // one issue
(Ch 83)
    }
}

class RedactorTest {
    @Test fun `scrubs PII from error context`() {
        val safe = redactor.redact(eventWith("user@example.com"))
        assertFalse(safe.context.values.any { it.contains("@") })   // Ch 28
    }
}

class SeverityRouterTest {
    @Test fun `fatal pages, transient samples`() {
        assertEquals(Route.PAGE, router.route(fatalEvent())); assertEquals(Route.SAMPLE,
router.route(transientEvent()))
    }
}
```

What to test: classification (transient/permanent/domain/severity, Ch 67); stable fingerprinting (type+location groups, Ch 83); severity routing (page/dashboard/sample, Ch 83); PII redaction (Ch 28); consent + severity sampling (Ch 82); durable buffering survives crash (Ch 83).

87.15 Production Observability

- **Metrics:** error **volume by severity/domain/fingerprint**, **top issues** (by count/users/ velocity, Ch 83), **crash-free rate** (Ch 83), transient vs permanent ratio, **new-in-release** issues, pipeline health (buffer/upload/sampling, Ch 82).
- **Logs:** structured telemetry events — **PII-redacted** (Ch 28).

- **Debugging:** error flood → check fingerprinting/dedup (Ch 83); false pages → check transient classification (Ch 67); un-investigable error → check correlation ids (Ch 84); duplicate issues → check fingerprint stability (Ch 83); PII leak → check redaction (Ch 28).
- **Alerting: fatal + new-in-release + high-impact** (page, Ch 83); **velocity spike** on any fingerprint (regression); crash-free-rate drop (Ch 83); pipeline drop (Ch 82).
- **Dashboards:** top issues by severity/domain/velocity; crash-free trend (Ch 83); transient/ permanent split; correlation drill-down to trace (Ch 84).

87.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Telemetry pillars?* Logs, metrics (Ch 85), traces (Ch 84), events (Ch 82), errors (Ch 83).
2. *Why unify?* One correlated story across pillars (Ch 84).
3. *Why classify errors?* Turn noise into actionable, routed signal.
4. *Classification axes?* Fatal/non-fatal, transient/permanent, domain.
5. *Why fingerprint?* Group duplicates into one counted issue (Ch 83).

Classification

6. *Fatal vs non-fatal?* Crash (Ch 83) vs handled (Ch 23).
7. *Transient vs permanent?* Retryable (timeout/503, Ch 67) vs bug (400/parse).
8. *Why it matters?* Drives handling (retry vs report) + severity.
9. *Domain?* Network/db/ui/payment — routing + ownership.

Fingerprinting

10. *How?* Type + top app frame/location (not raw message).
11. *Why not message?* Dynamic values make each unique → 10k issues.
12. *What's kept?* Count, first/last seen, users, versions (Ch 83).

Severity/routing

13. *Rank by?* Impact: fatal + volume + velocity + new-in-release.
14. *Route?* Page (fatal/high-impact), dashboard (rare), sample (transient).
15. *Sampling?* Keep all fatals, sample transients (Ch 82).
16. *Velocity?* Spiking matters as much as volume.

Correlation/privacy

17. *Correlate?* Trace/correlation/session ids (Ch 84).
18. *Why?* Link error to trace/metrics/context.
19. *PII?* Redact at boundary — errors are the top leak (Ch 28).
20. *Consent?* Follows the analytics regime (Ch 82).

Reliability

21. *Crash report survives?* Durable buffer, persist-before-death (Ch 83).
22. *Cost?* Batch + severity-sample (Ch 82).

Testing/Production

23. *Test classification?* Transient/permanent/severity cases.
24. *Alert fatigue?* Severity routing — page fatals only.
25. *Error flood?* Fingerprint/dedup + sampling.
26. *Un-investigable?* Correlation ids (Ch 84).

Design-change challenges

1. **Tame a 50k-errors/day flood.** Fingerprint/dedup + severity routing + sampling (Ch 83/82).
2. **Stop paging on network blips.** Classify transient → low severity/sample (Ch 67).
3. **Connect a slow trace to its error.** Correlation ids (Ch 84).
4. **Prevent PII in error reports.** Redact at the boundary (Ch 28).
5. **Group a bug reported 10k times.** Stable type+location fingerprint (Ch 83).

Mock walkthrough (abridged)

Clarify (which signals, severity routing, PII?) → unified pipeline ingesting errors/logs/ metrics/traces/events (Ch 82/83/84/85) → classify (fatal/non-fatal, transient/permanent, domain) → fingerprint/dedup (Ch 83) → severity + velocity routing (page/dashboard/sample) → correlate ids (Ch 84) → redact PII + consent (Ch 28) → durable buffer + severity sample (Ch 82) → observability (top issues, crash-free, Ch 83).

87.17 Practical Exercises

1. **Beginner.** Classify errors as transient vs permanent (Ch 67) and route severity (page fatal, sample transient). *Eval:* correct classification + routing.
2. **Intermediate.** Implement stable fingerprinting (type + location) so duplicate errors group into one issue (Ch 83). *Eval:* same bug → one fingerprint.
3. **Advanced.** Add PII redaction at the boundary (Ch 28), correlation ids (Ch 84), and severity sampling (keep fatals, Ch 82). *Eval:* PII-safe; correlated; cost-bounded.
4. **Production debugging.** Investigate an error flood (fix dedup) and false pages (fix transient classification). *Eval:* flood tamed; pages focused.
5. **System design challenge.** Design a unified telemetry pipeline correlating logs/metrics/ traces/errors/events (Ch 84) with severity routing and cost governance. *Eval:* one story; actionable; affordable.

(Solutions withheld unless requested.)

87.18 Chapter Summary & Reference Sheet

Key decisions: build a **unified telemetry pipeline** that ingests **all observability pillars** — logs, **metrics** (Ch 85), **traces** (Ch 84), **analytics events** (Ch 82), and **errors/crashes** (Ch 83) — through one **reliable, batched, consented, PII-safe** path, and **classify every error** along three axes: **fatal vs non-fatal** (crash, Ch 83, vs handled, Ch 23), **transient/retryable vs permanent/bug** (timeout/503, Ch 67, vs 400/parse), and by **domain** (network/db/ui/payment) — so classification **drives both handling** (retry vs report, Ch 23/67) **and severity**; **fingerprint and deduplicate** errors by a **stable signature** (exception type + top app frame/location, **not** raw message) so thousands of occurrences collapse into **one counted, ranked issue** (Ch 83); **rank by severity and velocity** and **route** accordingly (**page** the fatal, new-in-release, high-impact ones; **dashboard** the rest; **sample** expected transients, Ch 82); **correlate** every signal by **trace/correlation/session ids** (Ch 84) so an error links to the trace, metrics, and context that explain it; and **redact PII at the boundary** with consent (Ch 28) while **durably buffering** (so the crash's own report survives, Ch 83) and **severity-sampling** for reliability and cost (Ch 82) — turning a flood of raw failures into **actionable, deduplicated, correlated, PII-safe** signal.

Concepts: telemetry pillars unified (Ch 82/83/84/85), error classification (fatal/non-fatal, transient/permanent, domain), fingerprinting/dedup (Ch 83), severity + velocity routing (page/ dashboard/sample), correlation ids (Ch 84), PII redaction + consent (Ch 28), durable buffering

- severity sampling (Ch 82), classification-drives-handling (Ch 23/67).

Common mistakes: siloed pillars (no correlation); ungrouped errors (flood); fingerprinting by raw message (duplicate issues); paging on transients (alert fatigue); no severity sampling (cost/noise); PII in error text; losing the crash's own report; unclassified errors (can't route or handle).

Checklist:

- [ ] unified pipeline ingesting errors/logs/metrics/traces/events (Ch 82/83/84/85)
- [ ] classify fatal/non-fatal + transient/permanent + domain
- [ ] fingerprint/dedup by type+location (Ch 83)
- [ ] rank by severity + velocity
- [ ] route page/dashboard/sample
- [ ] correlate via ids (Ch 84)
- [ ] redact PII + consent (Ch 28)
- [ ] durable buffer (crash survives, Ch 83)
- [ ] severity sampling (keep fatals, Ch 82)
- [ ] classification drives handling (Ch 23/67).

Key APIs/concepts: Crashlytics fatal + non-fatal (`recordException` , Ch 83), custom keys/ fingerprints (Ch 83), structured telemetry events, SHA-256 signature fingerprinting, severity/ velocity routing (backend), correlation ids (Ch 84), PII redaction (Ch 28), reliable batched pipeline (Ch 82), transient classification (Ch 67).

References:

- Observability pillars (logs/metrics/traces) — observability literature
- Crashlytics issue grouping/fingerprinting: https://firebase.google.com/docs/crashlytics
- OpenTelemetry (unified signals + correlation): https://opentelemetry.io

- (composes) Ch 82 Analytics, Ch 83 Crash/ANR, Ch 84 Tracing, Ch 85 Perf, Ch 28 Privacy, Ch 23 Error Handling, Ch 67 Network Errors

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/transient/failure diagrams; classifier + fingerprinter + router + pipeline code; concurrency (dedup+sampling, classify- drives-handling, durable buffer); failure (classify+fingerprint+route+redact principle); 5 levels; unified-vs-siloed, fingerprint-key, severity-sampling trade-offs; security (redact, consent, opaque ids); perf (cheap classify, severity sample, dedup); classifier+fingerprint+ router+redactor tests; observability (top issues, crash-free, Ch 83); 26 Q&A + 5 challenges. ✅
- **Expand later:** crash/ANR (Ch 83), tracing (Ch 84), analytics (Ch 82), privacy (Ch 28), error handling (Ch 23/67).
- **Interview coverage:** exceeds ≥25 bar.